

Report 2: Q52.

Smart Public Transport Scheduling Optimization (Greedy + Graph Algorithms)

Title

Smart Public Transport Scheduling Optimization Using Greedy and Graph Algorithms

Abstract

Efficient public transportation is essential for reducing traffic congestion and improving urban mobility. This project investigates the use of Greedy and Graph Algorithms to optimize bus and train scheduling. Transportation networks are modeled as weighted graphs, where stations and stops represent vertices and routes represent edges. Dijkstra's algorithm is applied to determine shortest travel paths, while Greedy scheduling prioritizes high-demand routes. The proposed approach minimizes passenger waiting time, improves route coverage, and enhances overall transportation efficiency.

Problem Statement

A smart city transportation system must efficiently schedule buses and trains to minimize passenger waiting time and maximize route coverage. Due to increasing urban population and traffic congestion, intelligent scheduling strategies are required. Greedy and graph algorithms can optimize route allocation and timetable generation.

Analysis

Graph Representation

Transportation network is represented as a graph:

1. Vertices:
 - a. Bus Stops

- b. Railway Stations
2. Edges:
 - a. Routes connecting locations

Edge weights represent:

1. Travel time
2. Distance
3. Passenger demand

Algorithms Used

1. Greedy Algorithm

Used for:

- Bus allocation
- Route prioritization
- Peak-hour scheduling

Greedy strategy:

- Select route with highest demand first.
- Allocate vehicles accordingly.

2. Graph Algorithms

1. Dijkstra's Algorithm

Finds shortest travel paths.

2. Minimum Spanning Tree (MST)

Reduces infrastructure cost and route redundancy.

3. Breadth First Search (BFS)

Ensures network connectivity and coverage.

Constraints

1. Route Overlap

Multiple routes may serve identical regions.

Problems:

1. Resource wastage
2. Increased operating cost

2. Passenger Demand Variability

Demand changes:

1. Morning peak
2. Evening peak
3. Weekends

Scheduling must adapt dynamically.

3. Limited Fleet Size

Limited number of buses and trains.

Efficient allocation is essential.

4. Traffic Conditions

Road congestion affects bus schedules.

Real-time adjustments may be required.

Efficient Scheduling Solution

Step 1: Demand Collection

Gather passenger demand data.

Step 2: Network Graph Creation

Create graph representing transportation routes.

Step 3: Priority Assignment

Use greedy strategy:

- High-demand routes receive priority.

Step 4: Route Optimization

Apply Dijkstra's algorithm to determine shortest routes.

Step 5: Schedule Generation

Allocate buses and trains based on demand and route efficiency.

Step 6: Continuous Monitoring

Update schedules using real-time passenger and traffic data.

Scalability in Large Networks

Large cities may contain:

- Thousands of stops
- Hundreds of routes
- Millions of daily passengers

Scalability Techniques

Distributed Processing

Use cloud-based systems.

Parallel Route Computation

Optimize multiple routes simultaneously.

Dynamic Scheduling

Adjust schedules based on live traffic and demand.

Performance Efficiency

Let:

- V = number of stations/stops
- E = number of routes

Greedy Route Selection

Time Complexity: $O(E \log E)$

Dijkstra Algorithm

Time Complexity: $O((V + E) \log V)$

Schedule Generation

Time Complexity: $O(V + E)$

Overall system remains efficient for large transportation networks.

Results Interpretation

Expected Benefits

Metric	Improvement
Passenger Waiting Time	↓
Route Coverage	↑
Vehicle Utilization	↑
Fuel Consumption	↓
Operational Cost	↓

Urban Mobility Impact

The optimized scheduling system provides:

- Faster travel
- Better connectivity
- Reduced congestion
- Improved public transport reliability
- Increased passenger satisfaction

Conclusion

Greedy and graph algorithms provide an effective framework for smart public transport scheduling. By prioritizing high-demand routes and optimizing travel paths through graph-based techniques, cities can significantly reduce waiting times, improve coverage, and enhance urban mobility. The approach is scalable, computationally efficient, and suitable for modern smart-city transportation systems.